

Folder 02: Study Selection (PRISMA Process)

1. Overview

This folder contains the documentation for the multi-stage screening and selection process of our Systematic Literature Review (SLR) titled: " *Datasets and Data Preprocessing Techniques for Machine Learning-Based Insider Threat Detection*." The process followed the PRISMA 2020 (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines to ensure transparency, reproducibility, and the minimization of selection bias.

2. Selection Methodology

The selection process was conducted in four distinct stages to filter for technical depth and relevance:

- **Stage 1: Identification:** Initial database searches across Web of Science, Scopus, IEEE Xplore, ScienceDirect, and ACM resulted in **639 raw records**.
- **Stage 2: Initial Filtration:** Automated and manual filters were applied to include only peer-reviewed journal articles published between **2021 and 2025**, narrowing the pool to **397 records**.
- **Stage 3: Screening (Title & Abstract):** Utilizing **Zotero with Rayyan Ai** for deduplication and metadata management, titles and abstracts were screened for relevance to insider threat data transformation. This stage resulted in **117 articles** found eligible for full-text review.
- **Stage 4: Eligibility Assessment (Full-Text):** A rigorous full-text review of the 117 articles was conducted against our inclusion/exclusion criteria and Quality Assessment (QA) framework. This final stage resulted in the inclusion of **110 primary studies**.

3. Document Descriptions

File Name	Description
PRISMA_2020_Flow_Diagram.pdf	A high-resolution visual mapping of the record counts, deduplication, and exclusion numbers at each phase of the SLR.
Selection_Criteria_Notes.pdf	Detailed definitions of the inclusion/exclusion criteria, specifically how we defined "Feature Engineering" vs. "General ML" for the purpose of this study.